

CY 203P : Inorganic Chemistry Laboratory

Course Code : CY 203P

Course Name : Inorganic Chemistry Laboratory

L-T-P-C : 0-0-4-2

Intended for : BS Chemical Sciences

Prerequisite : None

Mutual Exclusion: None

Approval: 42nd Senate

Course Contents

1. General introduction to inorganic laboratory
2. Basic concepts of quantitative analysis
3. Errors in chemical analysis data
4. Qualitative analysis: Inorganic semi micro qualitative analysis with four radicals
5. Quantitative analysis:
 - (a) Volumetric Analysis:
 - i. Acid-base titrations relevant to the neutralizing power of antacids
 - ii. Complexometric and spectroscopic estimation of metal ions
 - b) Gravimetric Analysis: • Estimation of barium/sulphate as barium sulphate
 - iii. Estimation of iron as ferric oxide etc.
6. Synthesis:
 - (a) Preparation of potash alum from scrap aluminum
 - (b) Preparation of hexamine Ni(II) chloride
 - (c) Preparation of tetramine Cu(II) sulphate

Textbooks:

1. A.J. Elias, **A Collection of Interesting General Chemistry Experiments**, Revised edition, Universities Press Pvt. Ltd., 2007.

References:

1. **Vogel's Textbook of Quantitative Chemical Analysis**, 5th Edition, Orient Longman, 1989.
2. **Vogel's Textbook of Macro and Semimicro Qualitative Inorganic Analysis**, 5th Edition, Orient Longman, 1982.
3. Robert J. Angelici, **Synthesis and Technique in Inorganic Chemistry**, 2nd Edition, University Science Books, 1991.
4. Lab Manual and Instrument Manuals